

🌀 M002 · While Loop

Topic: Repeat Until a Wall · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

A while loop checks first, then moves. It stops at a wall.

1 · Trace It 🧠

```
while no wall ahead:  
    move forward
```

The wall is 4 steps away. How many times does the agent move?

2 · Real Code

```
while not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

`agent.detect(BLOCK, FORWARD)` is true when a wall is ahead.

Fill the blanks:

```
while not agent.detect(BLOCK, ____):  
    agent.____(FORWARD, 1)
```

3 · Fix the Bug 🐛

The agent should move until the wall. This is backwards.

```
while agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

Write the fix:

4 · Write It

Make the agent move forward until a wall, then turn right.

Use `while` , `agent.detect` , `agent.move` , `agent.turn` .

5 · Show Your Work 📷 🎥

Open your homework world. Use a while loop to move the agent down the straight maze to the wall.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.